BASIC ELECTRICAL ENGG
Module III

DC Generator

DC Motor

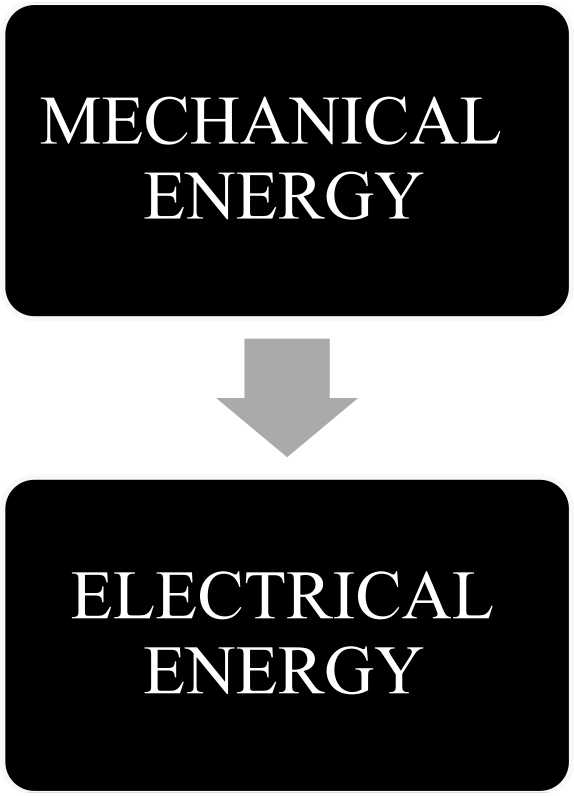
Classification

- Electrical Machines
  - DC Machines
  - AC Machines

DC GENERATOR

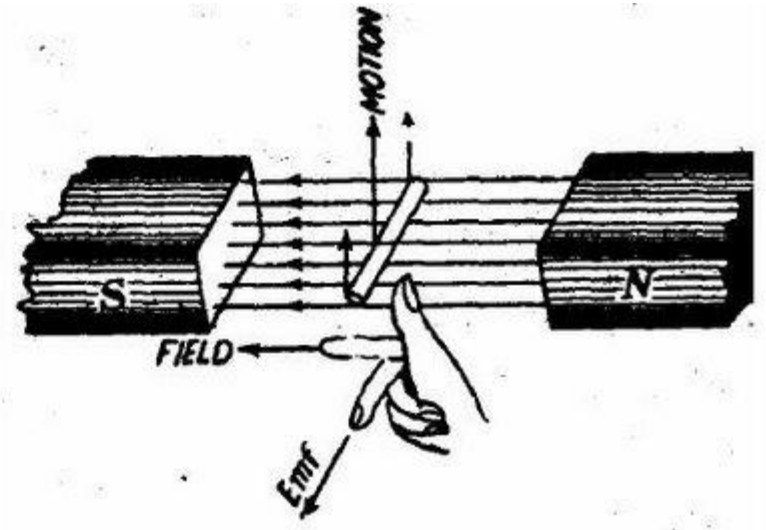
PRINCIPLE

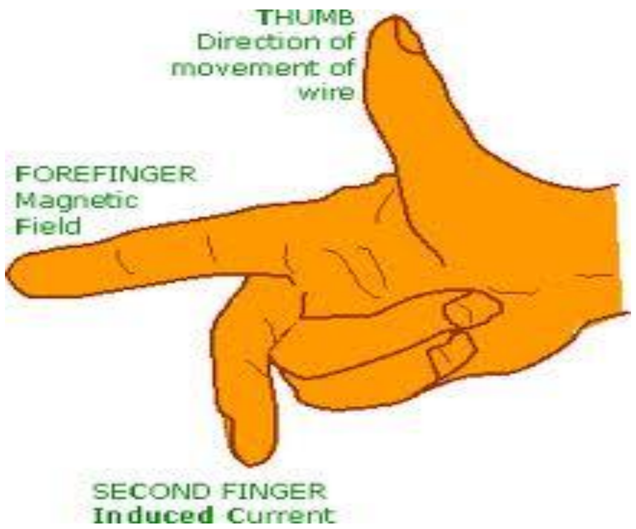
- The energy conversion in generator is based on the principle of the **production of dynamically induced emf.**
- Whenever a conductor cuts magnetic flux , dynamically induced emf is produced in it according **to Faraday's Laws of Electromagnetic induction**.
- This emf causes a current to flow if the conductor circuit is closed.



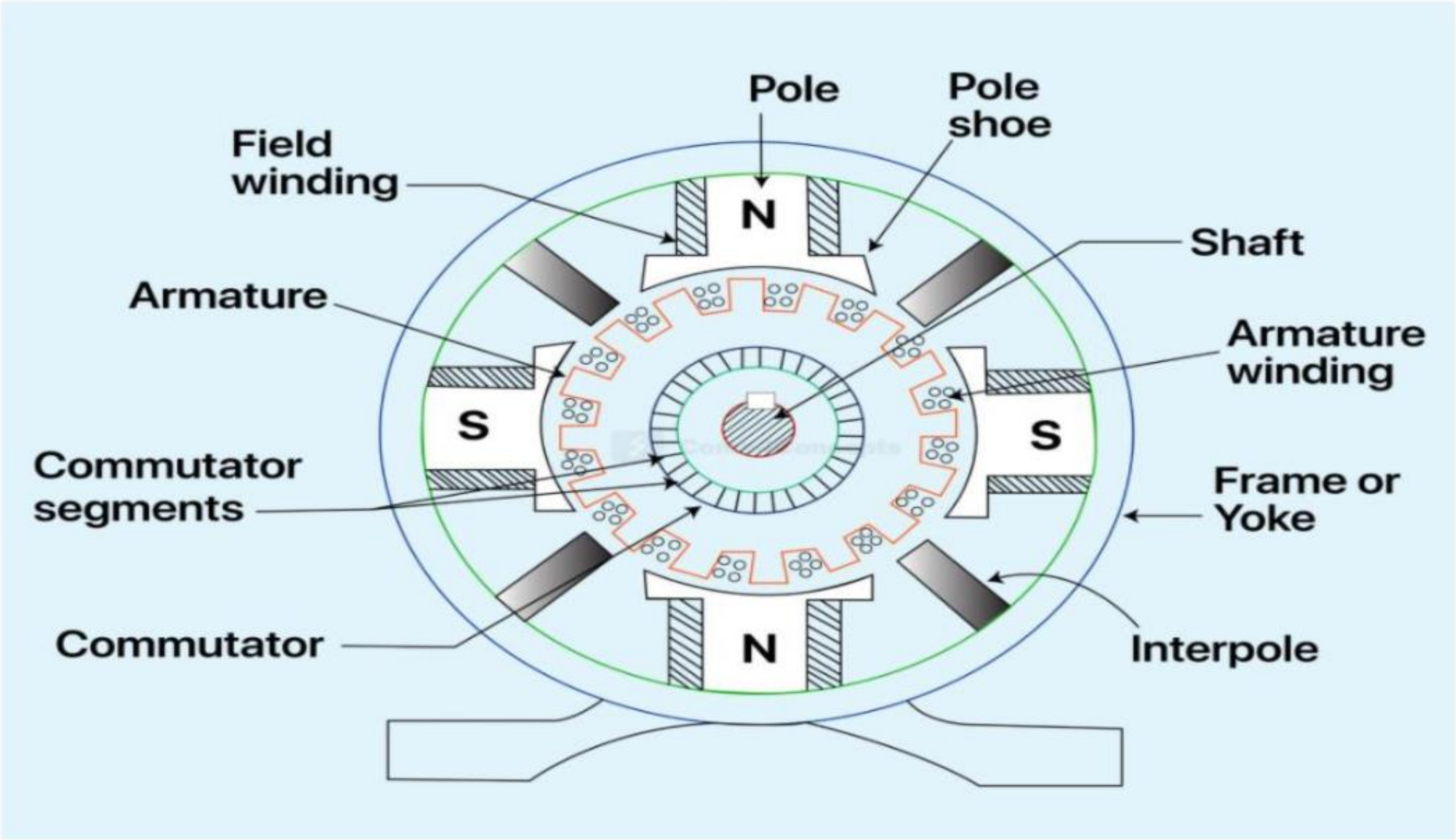
FLEMING’S RIGHT HAND RULE

- The direction of induced emf in the conductor is given by **<u>Fleming’s Right Hand Rule</u>**.
- First finger indicate direction of magnetic flux, Thumb indicate direction of motion and emf induced is indicated by Second finger.





DC GENERATOR - **Construction**



DC Generator

Construction

Yoke

- Outer frame made of cast iron and steel
- Provides Mechanical support to the generator
- Carries the magnetic flux that is generated

Pole core and pole shoes

- Field windings are connected in the pole core hence the pole core carries the field winding.
- Made up of cast iron or mild steel
- Pole shoes supports the windings wound on the pole core and also helps in distributing the generated flux uniformly into the machine

Interpoles

- Narrow poles fixed exactly midway between the main poles
- They are also called commutating poles.
- It helps in improving the commutation process

Field Winding

- Made up of copper
- To produce magnetic filed 2 types of magnets can be used Permanent magnet or electro magnet
- Electro magnets are usually used.

Armature

- Armature core is cylindrical type with slots
- It is laminated to reduce the eddy current
- Made up of steel
- Armature conductors are wound in this slots.

Armature Windings

- Copper coils are used
- The conductors are separately insulated
- Armature winding are of two types
    1. Lap winding
    2. Wave winding

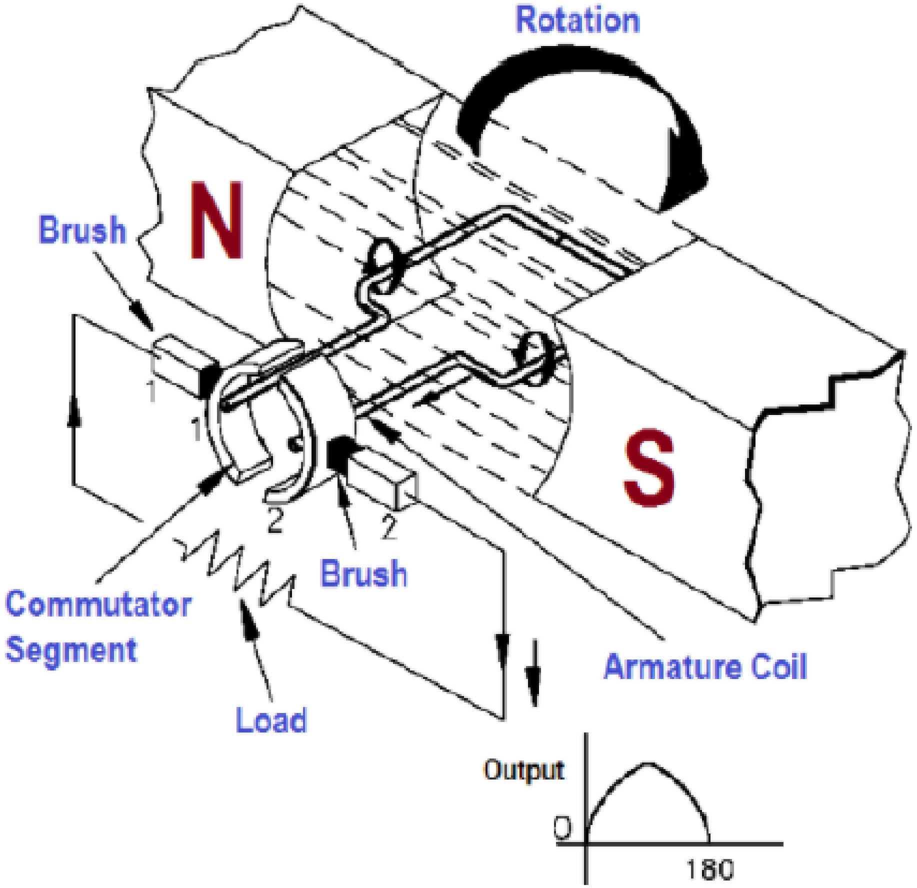
Commutator and brushes

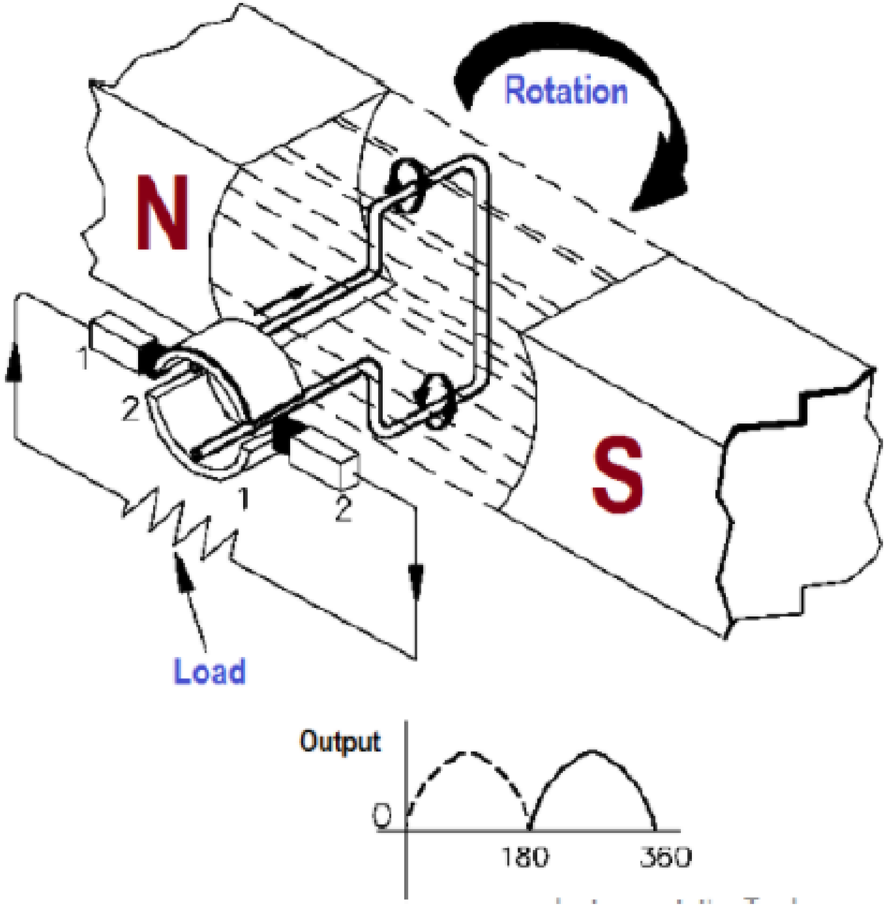
- Generator o/p is DC
- Commutator converts AC to DC
- Split rings are used as commutator
- Brushes are used to collect the generated current it is made up of carbon or graphite

Shaft

- The mechanical rod on which the armature rotates

Working


Rotation
Brush
N
S
1
1
2
2
Brush
Commutator
Segment
Load
Armature Coil
Output
0
180


Rotation
N
S
1
2
1
2
Load
Output
0
180
360

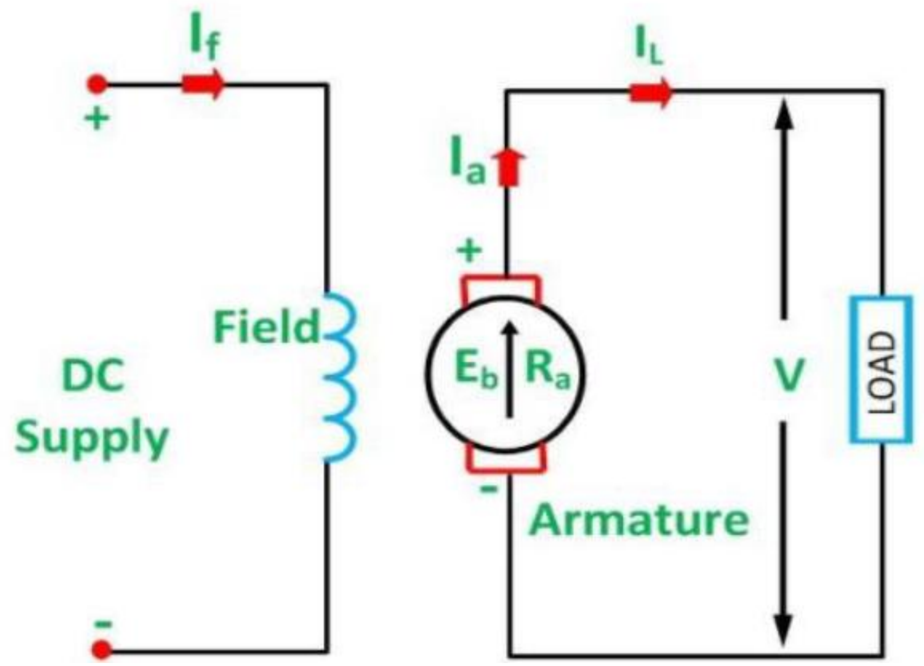
Types of DC Generator

Separately Excited

Self Excited

1) Series
2) Shunt
3) Compound

Separately Excited DC Generator



$$I_a = I_L = I$$

$$V = E_g - I_aR_a$$

$$\boxed{V = E_g - IR_a}$$

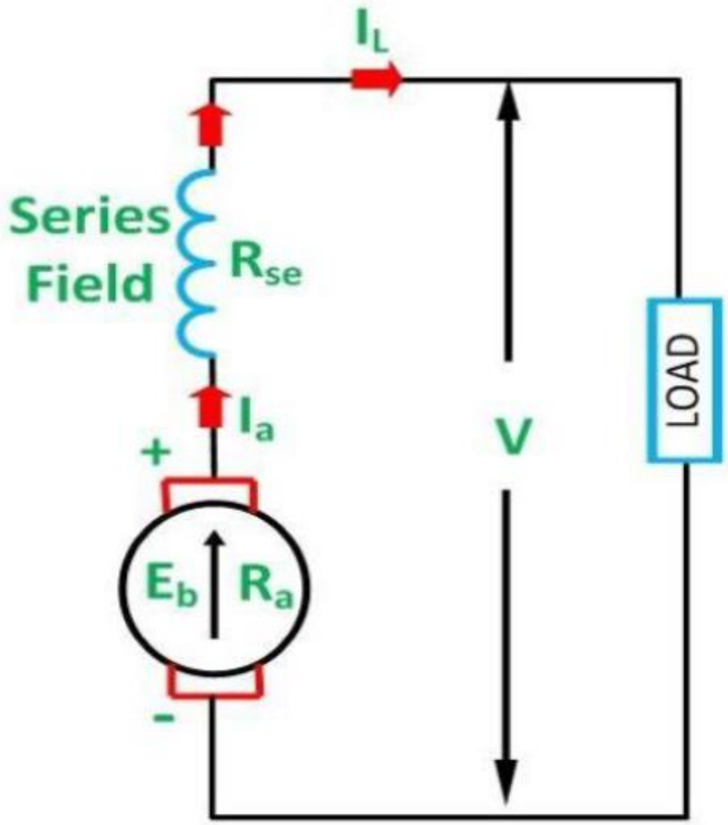
Self Excited DC Generator

Series

Shunt

Compound

Series Wound



Here,
$I_a = I_{se} = I_L = I$

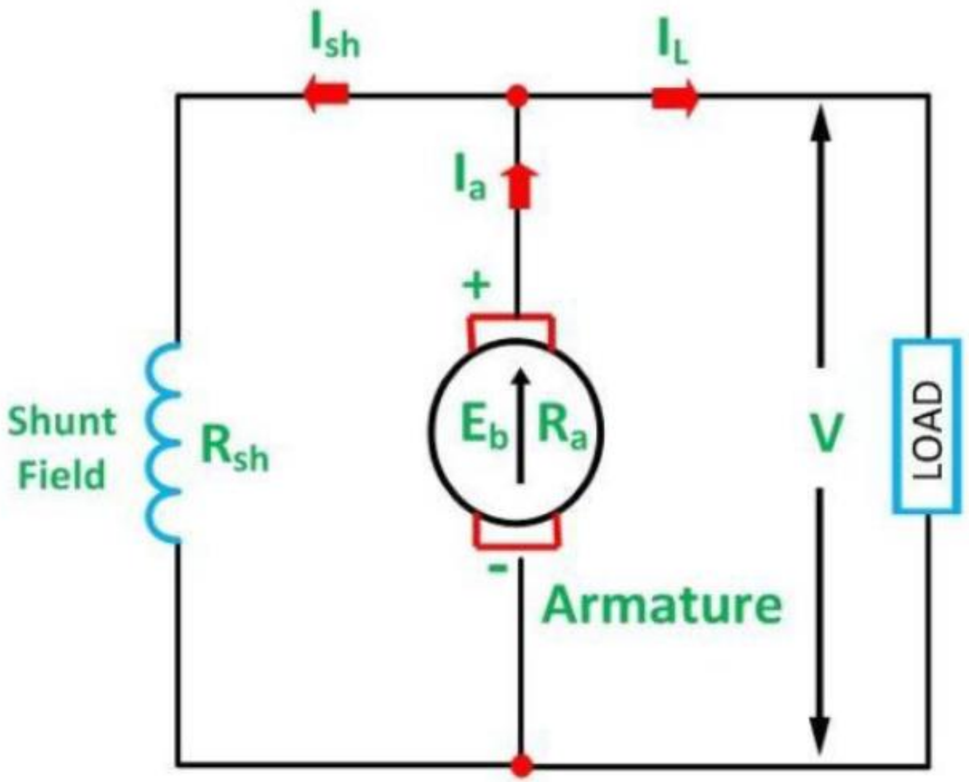
Considering drop across series winding,

$$V = E_g - I_aR_a - I_{se}R_{se}$$

So,

$$V = E_g - I\,(R_a + R_{se})$$

Shunt Wound



Here,

$I_a = I_{sh} + I_L$

And,

$I_{sh} = V/R_{sh}$

$V = E_g - I_aR_a$

So,

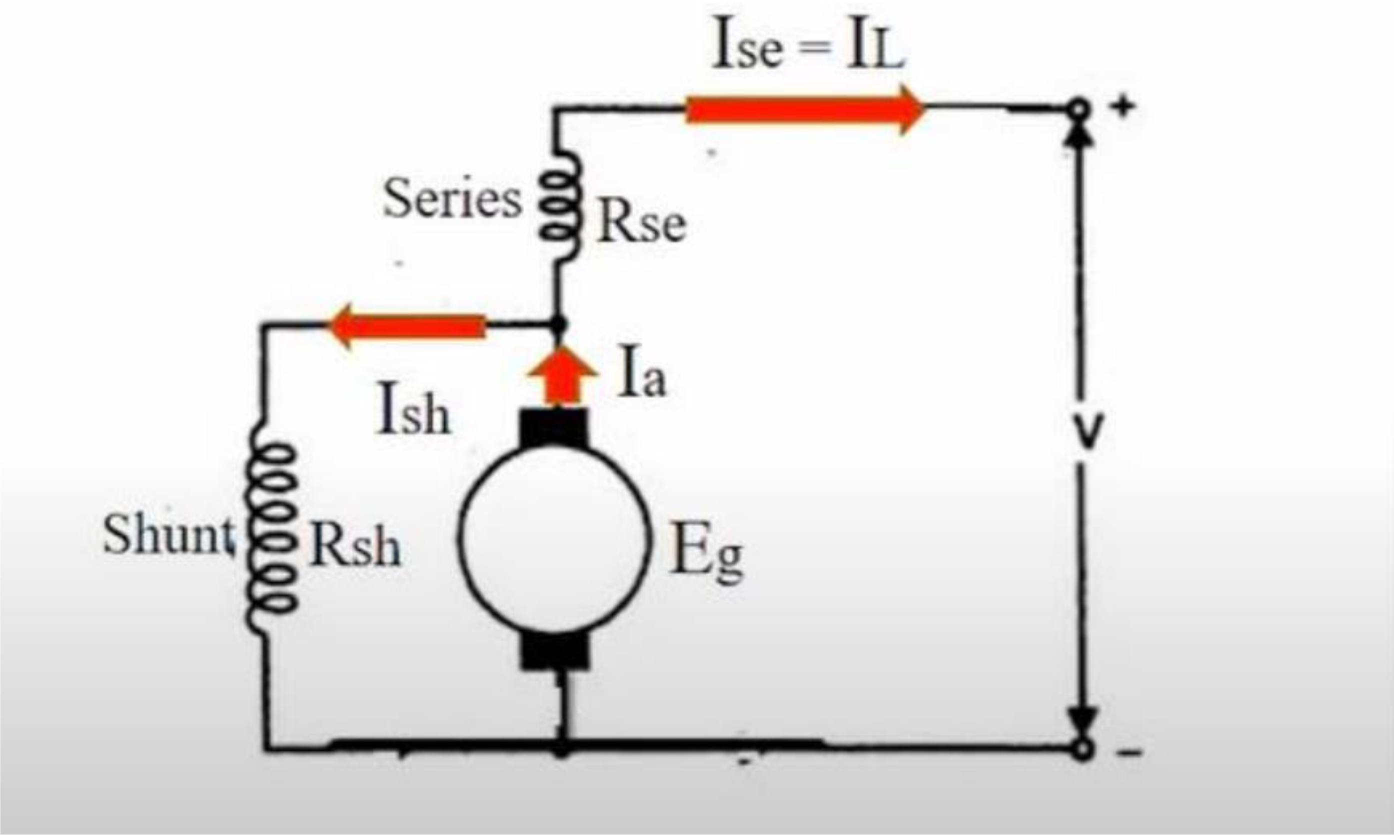
$\mathbf{V = E_g - (I_{sh} + I_L)R_a}$

Compound wound DC Generator

2 type

- Short shunt
- Long shunt

Short Shunt



$I_{se} = I_L$

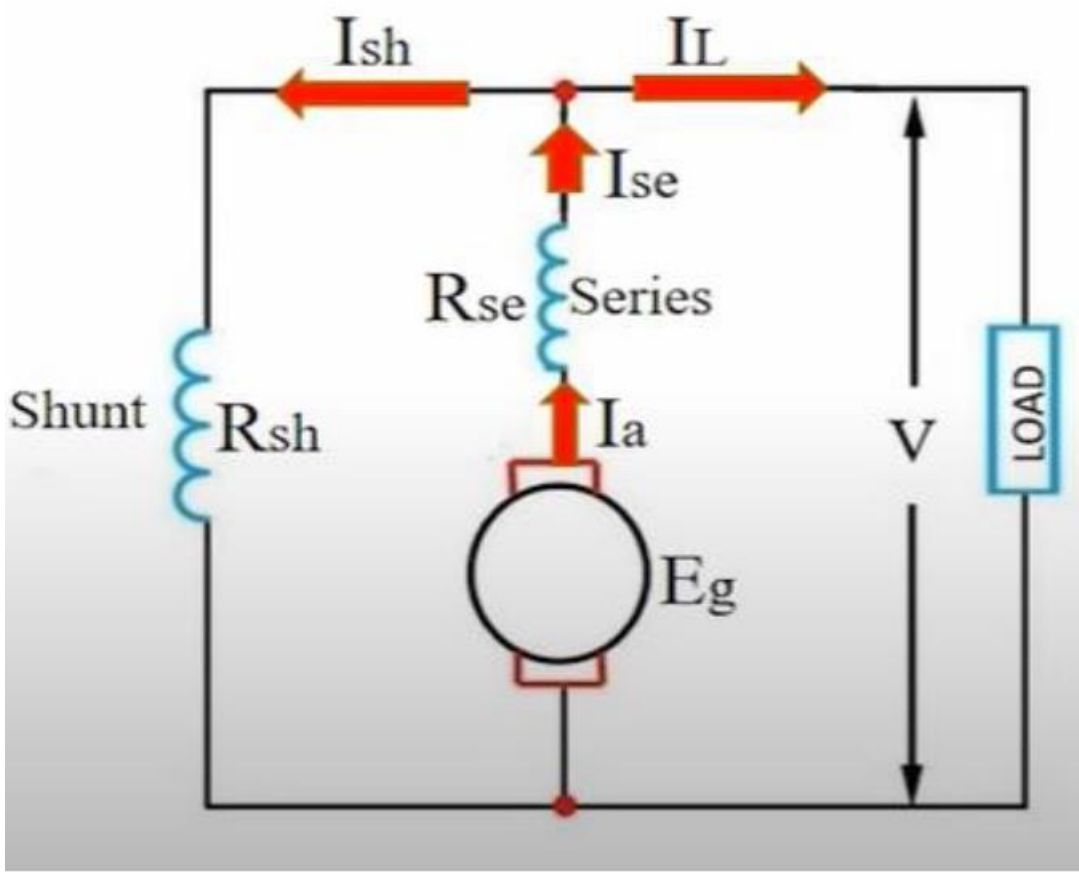
And, $I_a = I_{sh} + I_{se}$

$I_{sh} = (V + I_{se}R_{se})/R_{sh}$

So,

$$V = E_g - I_aR_a - I_{se}R_{se}$$

Long Shunt



$I_a = I_{se}$

$I_{sh} = V/R_{sh}$

Also,

$I_a = I_{se} = I_{sh} + I_L$

$V = E_g - I_aR_a - I_{se}R_{se}$

So,

$\mathbf{V = E_g - I_a (R_a + R_{se})}$

DC MOTOR

Converts electrical energy to mechanical energy

- Requires
  1. Field
  2. Conductor
  3. A current should be passed through the conductor

Constructional wise there is no difference between generator and motor.

- **When a current carrying conductor is placed in a magnetic field , it experiences a mechanical force whose direction is given by Flemings left hand rule and magnitude, $F = BIL$ newton**

- There is no basic difference between generator and motor in their constructional details

- The function of the commutator in the DC motor is the same as in case of a generator, by reversing current in each conductor as it passes from one pole to another, it helps to develop a continuous and unidirectional torque.

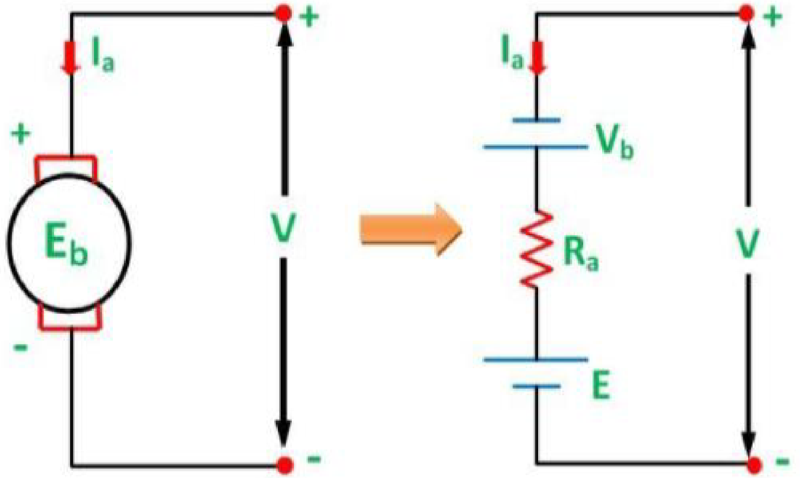
SIGNIFICANCE OF BACK EMF

When the armature rotates, the conductor also rotates and hence cut the flux.

EMF is induced in them whose direction was found by Flemings right hand rule , it is in opposition to the applied voltage

Because of its opposite direction it is termed as back emf E_b

$I_a = \frac{V - E_b}{R_a}$, R_a , is the armature resistance



$I_a \, R_a = V - E_b$

$V = E_b + I_a \, R_a$

Flux cut by one conductor = Pϕ

Time taken for one revolution = $\frac{60}{N}$

$$E_b = \frac{P\phi}{\frac{60}{N}} = \frac{P\phi N}{60}$$

$$E_b = \frac{P\phi N}{60} \times \frac{Z}{A}$$

Types of DC Motor

- Series
- Shunt
- Compound

V>Eb

DC Series Motor

Field winding is connected in series with the motor

$I_L = I_{SE} = I_a$

Terminal voltage $V = E_b + I_a R_a + I_{SE} R_{SE}$

$V = E_b + I_a R_a + I_{SE} R_{SE}$

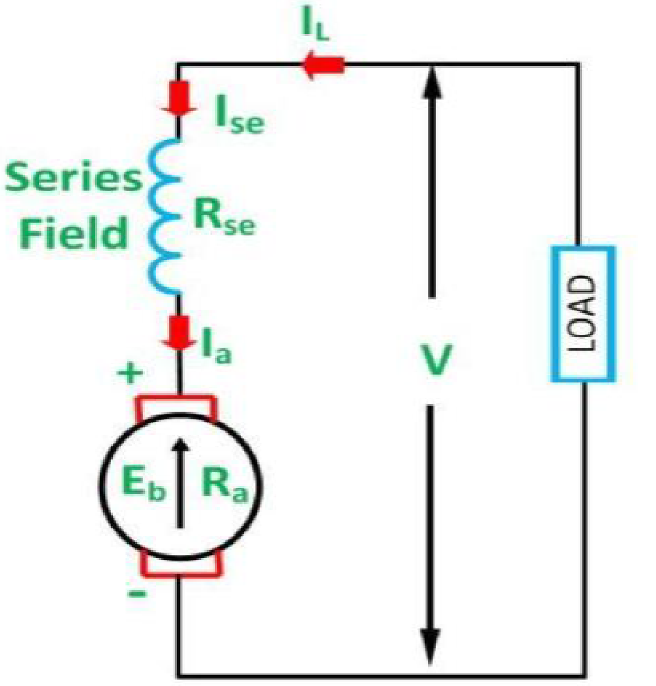
$V = E_b + I_a (R_a + R_{SE})$

$E_b = V - I_a (R_a + R_{SE})$

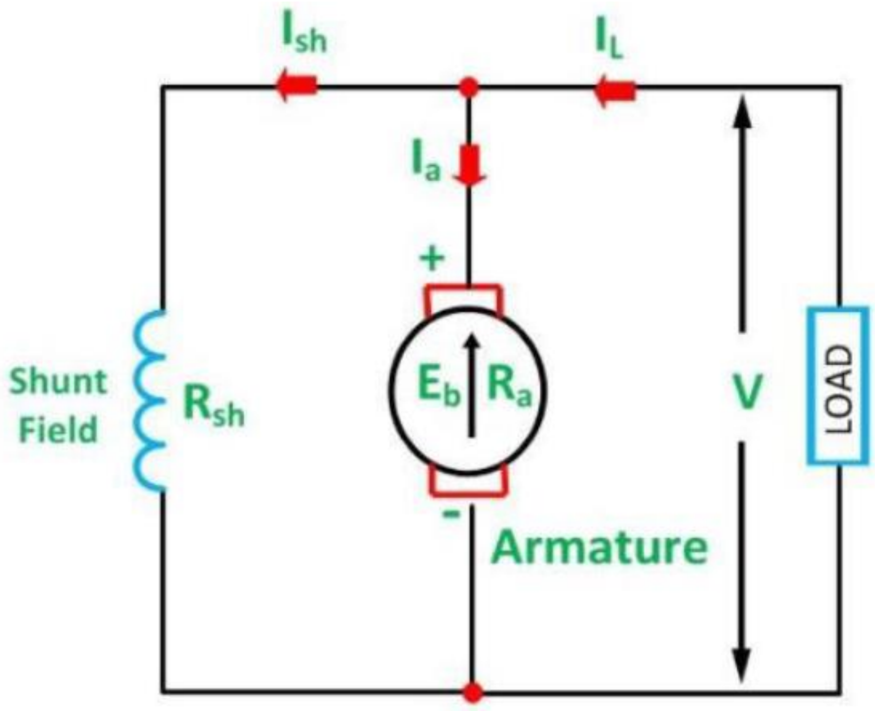
$E_b = \frac{P\phi N}{60} \times \frac{Z}{A}$

Z,A,P are constant

$V - I_a (R_a + R_{SE}) = \frac{P\phi N}{60} \times \frac{Z}{A}$



DC Shunt Motor



$I_L = I_a + I_{Sh}$

$I_a = I_L - I_{Sh}$

$I_{Sh} = \frac{V}{R_{sh}}$

$V = E_b + I_a\ R_a$

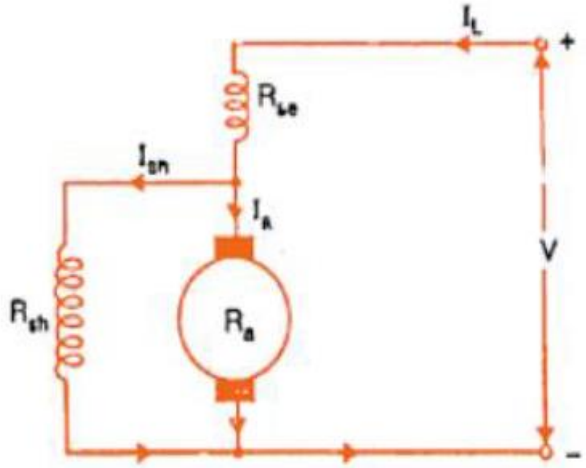
$V = E_b + (I_L - I_{Sh})\ R_a$

Hence, $(I_L - I_{Sh})\ R_a$

$E_b = V - (I_L - I_{Sh})\ R_a$

Compound Wound

Short Shunt



$I_L = I_{SE} = I_a + I_{Sh}$

$I_a = I_{SE} - I_{Sh}$

$I_{Sh} = \frac{(V - I_{SE} R_{SE})}{R_{sh}}$

$V = E_b + I_a\ R_a + I_{SE}\ R_{SE}$

$E_b = V - I_a\ R_a - I_{SE}\ R_{SE}$

Long Shunt

$I_L = I_{SE} + I_{Sh}$

$I_{SE} = I_a$

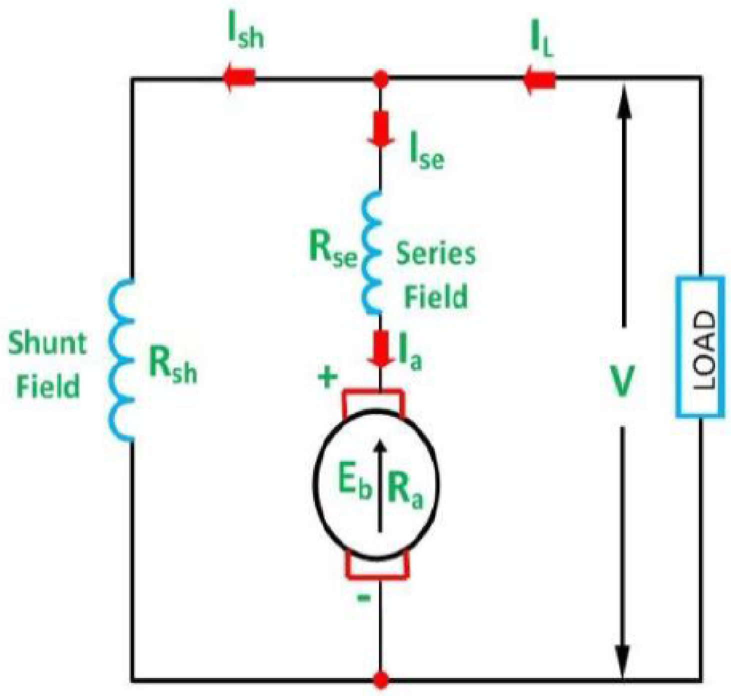
$I_L = I_a + I_{Sh}$

$I_{Sh} = \frac{V}{R_{sh}}$

$V = E_b + I_a R_a + I_{SE} R_{SE}$

$V = E_b + I_a(R_a + R_{SE})$

$E_b = V - I_a(R_a + R_{SE})$



Comparison between shunt, series and compound wound DC Motor

Type of motor	Characteristics	Applications
shunt	Approximately constant speed. Speed can be controlled. Medium starting torque (Upto 1.5 Full load torque)	For driving constant speed line shafting lathes, centrifugal pumps, machine tools, blowers and fans, reciprocating pumps.
Series	Variable speed. Speed can be controlled. High starting torque.	For traction work i.e. electric locomotives rapid transit systems trolley cars etc. cranes and hoists, conveyors
Cumulative compound	Variable speed Speed can be controlled. High starting torque.	For intermittent high torque loads, for shears and punches, elevators, conveyors, heavy planers, rolling mills, ice machines, printing presses, air compressors

APPLICATION

SHUNT GENERATOR

- Output is constant so, use for constant output purpose.
- For battery charging and lighting purpose.

SERIES GENERATOR

- Output vary with load.So, use as a voltage booster
- Use in electric trains.

CUMULATIVE COMPOUND GENERATOR

- For lightening and power services

DIFFERENTIAL COMPOUND GENERATOR

- For arc welding generator